

STEP 6

Model serving with FastAPI

Expose the model as a REST API with a strict input schema, versioned responses and request logging.

Endpoints

Method	Path	Purpose
GET	/health	Returns {"status": "ok"} when the model is loaded (503 otherwise). Used by Docker's HEALTHCHECK and the CI smoke test.
GET	/model-info	Model version, best model, scores, data hash, sklearn version
POST	/predict	One record in, one prediction out
POST	/predict/batch	A list of records in, a list of predictions out
GET	/docs	Interactive Swagger UI generated automatically

Run it

```
uvicorn api.main:app --reload          # then open http://localhost:8000/docs
```

Example request and response

```
POST /predict
{
  "Date": "2025-07-14", "Business_Type": "Restaurant", "Food_Category": "Prepared Food",
  "Purchase_Quantity": 180, "Shelf_Life_Days": 1, "Storage_Temperature": 24,
  "Daily_Demand": 110, "Promotion": "No", "Weather": "Rainy"
}

200 OK
{"waste_level": "High",
 "probabilities": {"Low": 0.0005, "Medium": 0.0031, "High": 0.9964},
 "model_version": "20261004-093839"}
```

Ordering 180 units against an expected demand of 110 gives **High** (99.6%). The same request with Purchase_Quantity 110 gives **Low** (95.6%).

Input validation

The request body is a pydantic model. Categories must come from the allowed lists, Purchase_Quantity must be > 0, Shelf_Life_Days ≥ 1, and Date must be a real date. Anything else is rejected with **HTTP 422** and a message naming the bad field, before the model runs. Shelf life, temperature, demand and weather are optional because the pipeline can fill them in.

Prediction logging

Every request is appended to logs/predictions.jsonl (timestamp, input, prediction). This log is the 'current data' that the monitoring step compares with the training data.

Why it matters

A service answer always says which model version produced it. When a prediction is questioned later, you can trace it back to the exact training run, data hash and scores in metrics.json.